Anower Hossen Zihad

zihadyeet@gmail.com zihadbappy.github.io Google Scholar linkedin.com/in/anower-zihad

Research interests: Distributed Deep Learning/AI, Beyond-Backpropagation Training, Continual Learning, Graph Neural Networks

EDUCATION

Montclair State University, MS Computer Science, GPA 4.00/4.00 — Expected Dec 2026

Chittagong Independent University, BS Computer Science & Engineering, GPA 3.90/4.00 — Sep 2018 – Aug 2022

RESEARCH EXPERIENCE

Montclair State University — Aug 2025 – Present

Research Assistant, School of Computing · supervisor: Chao Huang

1. Design and run experiments in **beyond-backpropagation** and **distributed deep learning**, implementing methods and training pipelines in **PyTorch** on university **HPC** systems and Google Colab.
2. Formulate research questions, analyze results, and write manuscripts with collaborators; presented at the ICCN Workshop, **IEEE INFOCOM 2026**, Tokyo, Japan.
3. Reported **50% lower** activation/gradient communication and up to **58% lower** peak memory versus standard **split learning** on CIFAR-10/100 and TinyImageNet, with comparable accuracy.
4. Extend split learning to **graph neural networks** in the beyond-backpropagation paradigm, benchmarking on Cora, Citeseer, and PubMed.
5. Apply **knowledge distillation** to mitigate forgetting in **continual** split learning, evaluating accuracy, convergence, runtime, communication, and memory.

PUBLICATIONS

1. **Decoupled Split Learning via Auxiliary Loss**. **Anower Zihad**, Felix Owino, Ming Tang, Chao Huang. ICCN Workshop, **IEEE INFOCOM 2026**, Tokyo, Japan. *arXiv:2601.19261*.
2. **Bangla Fake News Detection using Machine Learning, Deep Learning and Transformer Models**. Risul Islam Rasel, **Anower Hossen Zihad**, et al. **IEEE ICCIT 2022**, Cox's Bazar, Bangladesh, pp. 959–964.

UNDER REVIEW & IN PREPARATION

1. **Tackling Forgetting via Knowledge Distillation: A Case Study in Split Learning**. Under review, **IEEE ICNC 2027**.
2. **The Cost of Sequentiality in Zeroth-Order Split Learning**. Under review, **ICLR 2027**.
3. **Split Learning of GNN in a Beyond-Backpropagation Framework**. In preparation, **IEEE ICC 2027**.

INDUSTRY EXPERIENCE

TS4U LLC — Dec 2022 – Apr 2024

DevOps Engineer

1. Deployed and monitored production microservices on AWS with Docker and Kubernetes; managed cloud networking, compute, storage, databases, and access controls.
2. Automated infrastructure provisioning with Terraform and Ansible, reducing provisioning time by **30%**.
3. Implemented CI/CD workflows with GitLab CI/CD, Jenkins, and GitOps; integrated Prometheus, Grafana, and EFK monitoring to support troubleshooting.

TECHNICAL SKILLS

ML & scientific computing	PyTorch, TensorFlow/Keras, NumPy, pandas; GPU experimentation, HPC
Programming	Python, SQL, Bash, Java, JavaScript
Cloud & infrastructure	AWS (EC2, VPC, S3, RDS, IAM, ECR, EKS), Docker, Kubernetes, Terraform, Ansible
Delivery & monitoring	Git, GitLab CI/CD, Jenkins, GitOps, Prometheus, Grafana